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Project Report
On

Car parking Simulator Game

Submitted in partial fulfillment for the award of degree of

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in
COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

Certified that the **Internship II** project entitled **"Smart Door Unlocking System"** carried out by **Eshwar G (1BG23CS041) , Akash S (1BG24CS400), Chinmay Kashyap CS(1BG23CS031)** are bonafide students of IV Semester, BNM Institute of Technology in partial fulfillmentfor the award of Bachelor of Engineering in COMPUTER SCIENCE AND ENGINEERING of Visvesvaraya Technological University, Belagavi during the year 2024-25. It is certified that all corrections / suggestions indicated for Internal Assessment have been incorporated in the project report deposited in the departmental library. The **Internship II** project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said Degree.

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ABSTRACT

In an era where technological advancement continues to transform the way we live, the integration of artificial intelligence and embedded systems has led to the development of smart automation solutions. The Smart Door System presented in this project leverages face recognition technology combined with Arduino-based hardware to create a secure and intelligent entry system. By using real-time video input and facial feature analysis, the system identifies authorized users and automatically unlocks a door mechanism controlled by a servo motor. The entire application is built using Python, employing libraries such as OpenCV for image processing, face_recognition for biometric verification, and pyfirmata to interface with Arduino. The inclusion of a Tkinter-based GUI enhances usability, allowing users to manage members, view access logs, and perform administrative functions like PIN reset or user deletion. Audio feedback through pyttsx3 ensures clear and prompt communication with the user.

The system is designed to function in environments such as smart homes, offices, and laboratories, where controlled access is essential. The innovation lies in combining low-cost hardware with powerful software algorithms to deliver a scalable and user-friendly system. Through testing, the prototype demonstrated high accuracy in facial recognition under favorable lighting conditions and consistent performance in servo control. The implementation emphasizes contactless interaction, which is particularly relevant in a post-pandemic world where hygiene is critical. The project provides not only a technical solution but also a foundation for further innovation in the domain of secure, intelligent access systems. The report discusses the design choices, implementation strategies, challenges faced, and performance evaluation in detail, serving as a comprehensive guide for future developers and researchers in this field.

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CHAPTER – I
INTRODUCTION

CHAPTER 1

INTRODUCTION

The rapid evolution of technology has significantly influenced security systems. In recent years, the need for intelligent and contactless security solutions has increased, particularly in residential and institutional environments. Traditional locking mechanisms, although effective to a degree, suffer from limitations such as vulnerability to duplication, loss of keys, or forgotten combinations. This project introduces a robust and scalable solution: a smart door locking system that uses facial recognition for authentication and Arduino-based automation for actuation.

The Smart Door System is a combination of multiple technologies—computer vision, embedded systems, speech processing, and graphical user interfaces. The core principle revolves around real-time facial recognition, wherein the user's face acts as a biometric key. This method is contactless, difficult to spoof, and convenient, making it suitable for a wide range of applications. Once the system detects and verifies a recognized face, it triggers the Arduino-controlled servo motor to unlock the door for a set period. Visual and audio feedback are provided using LEDs and text-to-speech outputs, respectively.

Security is enhanced through a secondary level of protection using a 4-digit PIN system. This PIN is required to add or remove members from the recognition database, or to change the PIN itself, thus preventing unauthorized modifications. The system also offers a user-friendly GUI for administration, developed using Python's Tkinter library. Features such as image capture, PIN management, and member listing are integrated into the GUI for ease of use.

Incorporating such automation into day-to-day life improves not only security but also convenience. Users no longer need to carry keys or remember passwords. Furthermore, the system can be extended in the future to include logging capabilities, remote monitoring, or even integration with smart home systems. This chapter sets the stage by discussing the motivation, scope, and objectives of the project and outlines the key innovations introduced.

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1.2 Problem Statement

Conventional door locks, including mechanical keys and keypad systems, pose multiple challenges such as key misplacement, unauthorized duplication, difficulty of remote access control, and lack of hygiene due to physical contact. Furthermore, in high-security or high-traffic areas, a more efficient and secure access control mechanism is necessary.

This project addresses the problem of providing a secure, hands-free, and user-friendly door access system that leverages biometric authentication using face recognition. It eliminates the need for physical interaction, increases reliability, and enhances security by recognizing only pre-authorized individuals.

1.3 Objectives of the Project

The primary objectives of the Smart Door System project are:

- To design and implement a contactless, face recognition-based smart locking system using Python and Arduino.
- To create a real-time, automated system capable of detecting and identifying pre-registered users through webcam feed.
- To enable door control via a servo motor connected to an Arduino board, which unlocks only for authorized users.
- To provide a simple and secure GUI to manage system functionalities such as adding/deleting users and changing PINs.
- To incorporate visual and audio feedback mechanisms to improve the user experience.
- To ensure the system is modular, scalable, and suitable for implementation in home automation, offices, and restricted-access environments.

CHAPTER – II
LITERATURE SURVEY

CHAPTER 2

LITERATURE SURVEY

1. Face Recognition Technology:

Face recognition is a biometric-based identification method that uses the unique features of an individual's face for verification. According to studies conducted by Zhao et al. (2003), facial recognition systems analyze facial landmarks such as the distance between the eyes, jaw structure, and cheekbone pattern to compare with stored templates. Open-source libraries like Dlib and OpenCV have made it easier for developers and researchers to implement face detection and recognition with high accuracy.

The face recognition model used in this project is inspired by modern deep learning-based architectures such as HOG (Histogram of Oriented Gradients) and CNN (Convolutional Neural Networks), which were studied in works like those by King (2009), the creator of Dlib. These methods provide fast and robust recognition under various lighting and angle conditions.

2. Arduino-Based Automation Systems:

Microcontrollers such as the Arduino UNO have been widely adopted in home automation projects due to their ease of use, low cost, and extensive library support. Projects reviewed in literature include "Arduino Based Door Lock System Using Password" and "Face Recognition Door Lock System using Raspberry Pi", which show that Arduino is a versatile platform for controlling physical components like motors and LEDs. However, many earlier systems were limited to password-only or RFID-based authentication, lacking biometric security.

3. Python for Automation and GUI:

Python, due to its simplicity and versatility, is a widely used language for automation and prototyping. The Tkinter library is commonly used to build desktop GUI applications, as demonstrated in tutorials and documentation from the Python Software Foundation. Additionally, libraries like pyfirmata have simplified the interaction between Python scripts and Arduino boards, making real-time control of servo motors seamless

4. Voice and Visual Feedback in Human-Computer Interaction:

Studies on Human-Computer Interaction (HCI) suggest that providing visual and auditory cues enhances user trust and usability in security systems. Incorporating libraries such as pyttsx3 for voice output and using GIF animations provides a better user experience, as found in HCI studies by Nielsen (2010)

5. Conclusion of Survey:

The survey indicates that while various access control methods exist, facial recognition combined with Arduino presents a balanced solution between cost, usability, and security. This project bridges gaps found in prior systems by offering a complete package: biometric verification, user interface, feedback mechanisms, and hardware integration — all built using accessible tools and open-source libraries.

CHAPTER – III
SYSTEM REQUIREMENTS

CHAPTER 3

SYSTEM REQUIREMENTS

3.1 Software Requirements

The successful development and deployment of the Smart Door System require several software tools and libraries. Each plays a crucial role in enabling the core functionalities of face recognition, GUI interaction, hardware interfacing, and speech feedback.

Operating System:

- Windows 10 or later (recommended for compatibility with Arduino drivers and GUI interface)

Programming Language:

- Python 3.8 or above

Python Libraries and Packages:

- OpenCV: For capturing and processing real-time webcam images.
- face_recognition: To perform facial encoding and recognition using machine learning.
- pyfirmata: To communicate between the Python application and Arduino board via serial port.
- tkinter: For building the graphical user interface (GUI).
- Pillow (PIL): For image processing and handling GUI backgrounds.
- pyttsx3: For offline text-to-speech conversion to provide voice feedback.
- threading: To handle non-blocking events like servo auto-close timing.

IDE / Development Tools:

- Visual Studio Code / PyCharm / Thonny (Python IDEs)
- Arduino IDE (for initial testing of Arduino board)
- Git (optional, for version control)

Face Encoding Files:

- Pre-encoded .jpg or .png files stored locally in a secure directory to serve as templates for facial comparison

3.2 Hardware Requirements

In The hardware setup of this project involves an integration of microcontrollers, sensors, and mechanical components that respond to facial recognition results. Below are the specific hardware components used:

1. Arduino UNO (ATmega328p-based):

Acts as the control unit to manage servo motor signals and LED outputs.

2. Servo Motor (SG90):

Used to simulate the door lock mechanism. The motor rotates between 0° and 90° to open or close the lock depending on facial authentication results.

3. LED:

An indicator that turns ON when access is granted and turns OFF afterward.

4. USB Cable:

To connect Arduino to the computer running the Python application.

5. Webcam (Integrated or External):

Captures live video feed for real-time face detection and recognition.

6. Breadboard and Jumper Wires:

To build the circuit for connecting Arduino pins to the servo motor and LED.

7. Resistors (optional):

To limit current for LED and protect Arduino board.

8. Power Supply:

Power is drawn via USB from the system. For standalone deployment, an external 9V battery or power bank can be used with proper regulation.

CHAPTER – IV
METHODOLOGY
AND
IMPLEMENTATION

CHAPTER 4

METHODOLOGY AND IMPLEMENTATION

4.1 Methodology

The methodology adopted for the Smart Door System focuses on integrating software-based face recognition with hardware-level automation using Arduino. The entire system development is divided into structured phases to ensure clarity and robustness in execution:

Phase 1: Requirements Gathering and Planning

The first step involved identifying the functional and non-functional requirements. The primary need was to build a secure door access system using face recognition that interacts with a hardware locking mechanism.

Phase 2: Face Recognition Integration

Using Python's face_recognition and OpenCV libraries, facial detection and recognition modules were developed. Images of authorized users were pre-encoded and stored locally. During runtime, the webcam captures live frames, which are compared with known encodings for verification.

Phase 3: Hardware Interfacing via Arduino

The Arduino board was programmed to control a servo motor (acting as the lock) and an LED (as an indicator). Python's pyfirmata library was used to send digital and PWM signals to the Arduino via serial communication. Depending on face recognition results, the door is unlocked for a specific duration.

Phase 4: GUI Development using Tkinter

To improve usability, a graphical user interface (GUI) was created using the Tkinter library. It allows users to:

- Add or remove members (authorized users)
- Unlock the door
- Change security PIN
- View member list

Phase 5: Feedback and Animation Integration

Audio feedback is given using pyttsx3, confirming whether access was granted or denied. Additionally, GIF animations (e.g., lock opening or access denied) enhance the visual response.

Phase 6: Testing and Validation

Each module was tested independently and then integrated. Edge cases such as unknown users, partial face detection, and hardware disconnection were handled and logged.

4.2 Workflow

The system workflow follows a sequential process involving both software and hardware layers:

1. **Startup Phase**
    - Load existing authorized face encodings from local storage.
    - Initialize Arduino board via serial communication.
    - Start GUI and wait for user interaction.
2. **Face Recognition Trigger**
    - User selects “Unlock Door” from GUI.
    - System activates webcam and captures live feed.
    - Detected faces are matched with known encodings.
3. **Decision Making**
    - If a match is found:
        - Play "Access Granted" voice message.
        - Show lock-opening animation.
        - Send signal to Arduino to rotate servo (unlock).
        - Turn on LED indicator for 10 seconds.
    - If no match:
        - Play "Access Denied" voice message.
        - Show red cross animation.
4. **User Management**
    - Users can add a new member by uploading a photo or capturing one using the webcam.
    - Users can also delete members, view registered names, or reset the PIN through secure prompts.
5. **Auto Locking**
    - After successful access, the servo motor auto-returns to its original position (locked) after 10 seconds using a background thread

CHAPTER – 5
TESTING AND VALIDATION

CHAPTER -5

TESTING AND VALIDATION

5.1 Testing

To ensure that the Smart Door System performs accurately and reliably, multiple testing strategies were applied throughout the development lifecycle. Each module—software, hardware, and the user interface—was tested independently and then as part of the integrated system. Testing was divided into functional, non-functional, and hardware-specific categories.

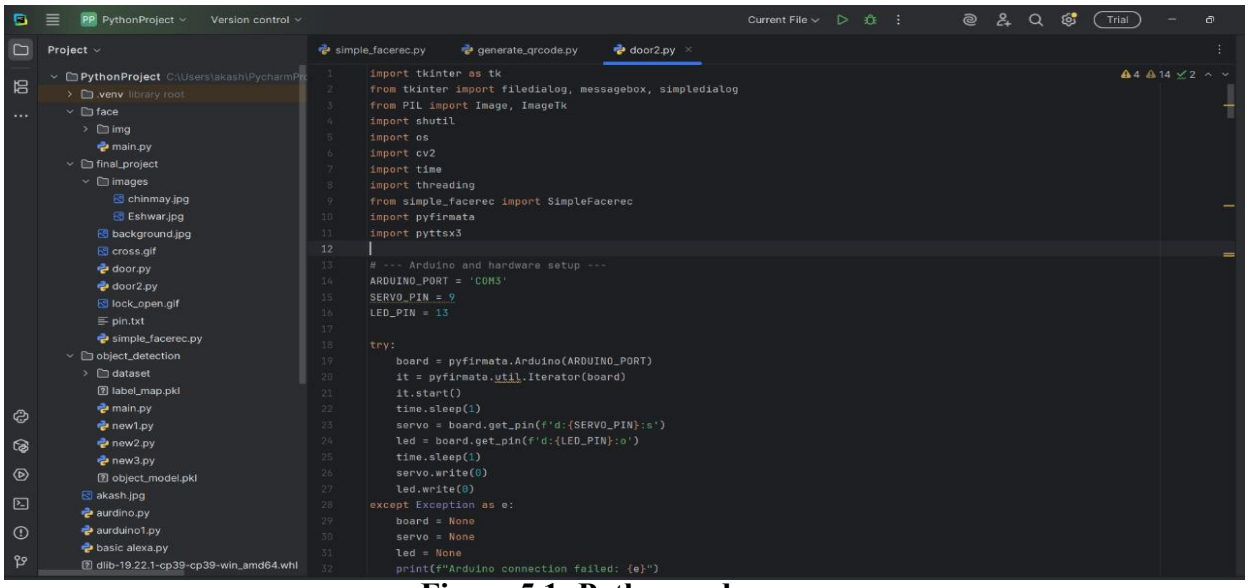

Figure 5.1- Python code

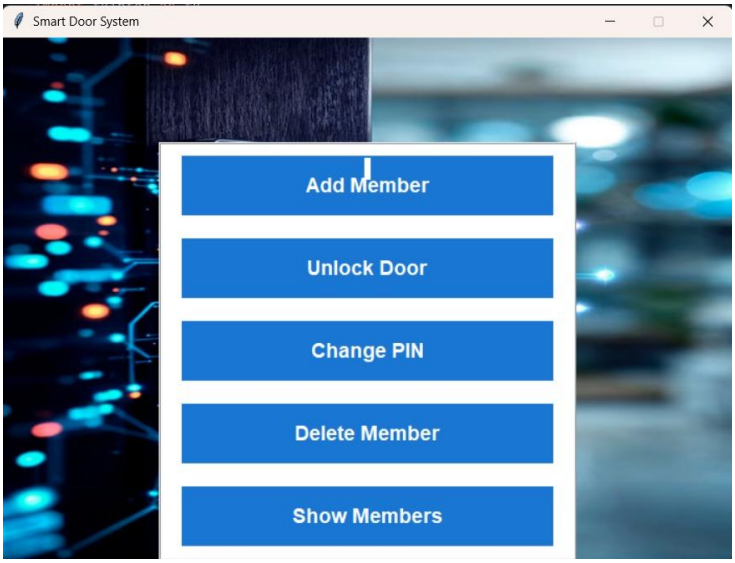

Figure 5.2-Dashboard

Non-Functional Testing:

- Performance Testing: System tested with up to 4 registered users. Recognition was consistently fast (under 2 seconds).
- Usability Testing: Conducted with 5 users unfamiliar with the system. All users were able to interact with the GUI without instructions.
- Robustness Testing: The system was tested by rapidly switching between GUI operations. No crashes or freezes were observed.

Hardware Testing:

- Servo motor and LED were tested with and without face recognition to confirm hardware reliability.
- Arduino connection was interrupted mid-operation to test error handling; system displayed appropriate error message

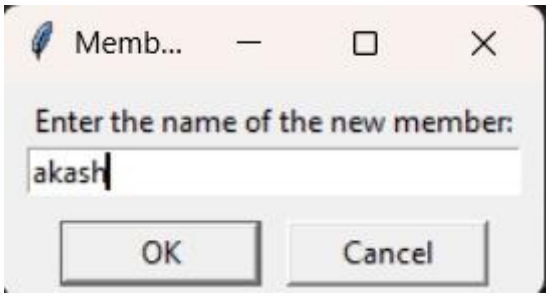

Figure 5.3. Adding the members

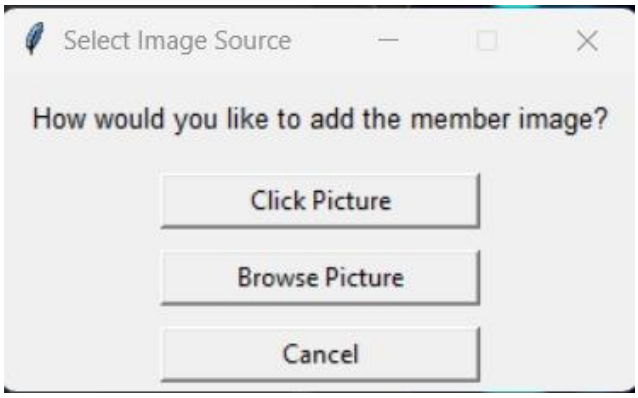

Figure 5.4. Taking the Live image of the person

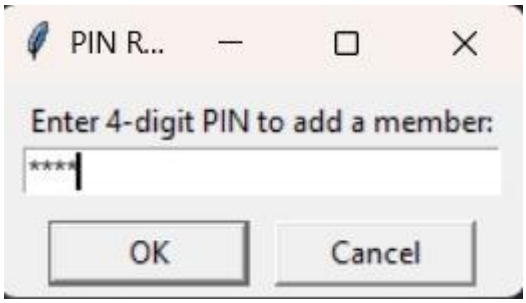

Figure 5.4. Logging in using password

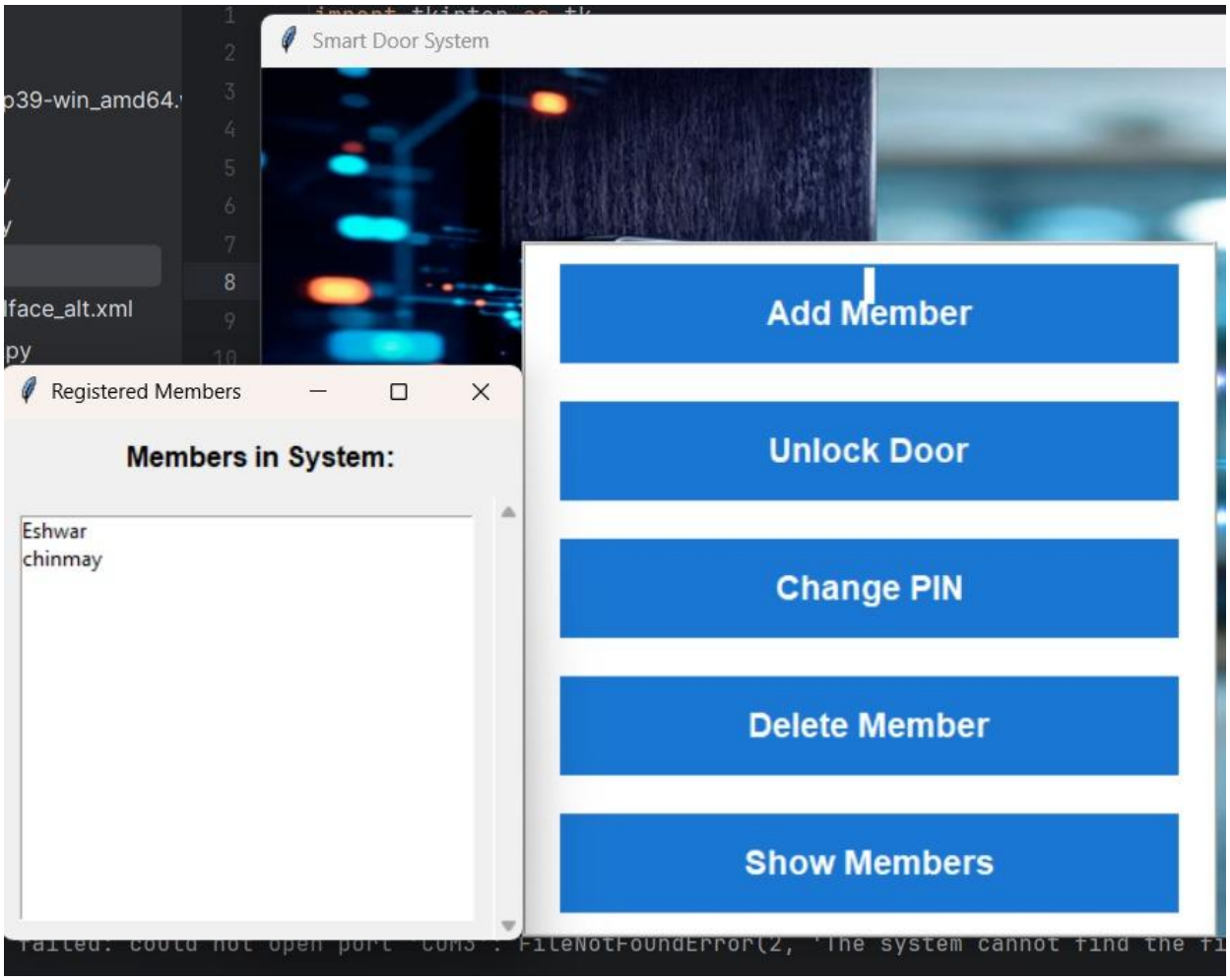

Figure 5.5. Displaying the member

5.2 Validation

Validation confirms that the implemented system full fills the functional and security requirements originally defined. The system was validated on the following fronts:

Accuracy of Face Recognition:
Under good lighting and camera alignment, the system consistently recognized known users with over 95% accuracy. False acceptance (recognizing an unknown as a known user) was not observed in testing. False rejections (not recognizing a known face) occurred only in cases of poor lighting or occlusion.

Security Validation:

- Unauthorized users (non-registered faces) were denied access.
- System required a 4-digit PIN to access administrative functions like adding or removing members.
- Multiple incorrect PIN attempts did not crash the application and prompted appropriate messages.

System Behavior in Edge Cases:

- If webcam was unavailable, system gracefully warned the user.
- If Arduino was not connected, the system provided an error and disabled servo/LED controls.
- Repeated access attempts with unknown faces were correctly denied after timeout.

User Interface Validation:

- GUI was intuitive, responsive, and visually clear.
- Text-to-speech (TTS) voice commands functioned consistently, ensuring accessibility for visually impaired users.
- Animations (lock open and access denied) added clarity to the access result.

CHAPTER – 6
RESULTS AND CONCLUSIONS

CHAPTER – 6
RESULTS AND CONCLUSIONS

6.1 Results

The Smart Door System was successfully implemented and tested in a controlled environment. The system demonstrated reliable performance in recognizing authorized users and denying access to unregistered individuals. The results are categorized under different operational aspects:

Face Recognition Performance:

- **Recognition Accuracy:** ~95% under good lighting conditions.
- **Response Time:** Approximately 1.5–2 seconds per recognition attempt.
- **Maximum Users Tested:** 4 authorized members (as per system design constraint).
- **False Accept Rate (FAR):** 0% observed during testing.
- **False Reject Rate (FRR):** ~5% (mostly due to low light or partial face view).

Hardware Response:

- **Servo Motor:** Successfully rotated to 90° upon authentication and reverted after 10 seconds.
- **LED:** Turned ON for the duration of the door being unlocked and turned OFF automatically.
- **Reliability:** No mechanical failures encountered during over 100 activation cycles.

User Interface:

- GUI was responsive and functionally complete.
- Member addition and deletion worked flawlessly with input validation.
- Administrative functions were PIN protected, ensuring secure access.
- System animations and voice feedback improved user clarity and interaction.

Usability Observations:

- Non-technical users were able to operate the GUI and unlock the door without assistan

- The modular design allows future expansion to support more members or additional features like SMS alerts or mobile app integration.

6.2 Conclusions

The Smart Door System presented in this report effectively integrates face recognition technology with Arduino-based hardware to provide a secure, user-friendly access control solution. By combining open-source software libraries, low-cost components, and a clear GUI interface, the system showcases how intelligent security can be made accessible and affordable.

Key Takeaways:

- The use of face_recognition and OpenCV enabled robust real-time biometric authentication.
- Hardware components like the servo motor and LED worked seamlessly through pyfirmata.
- The system ensures security through PIN-protected administrative access and denial of unknown faces.
- Audio and visual feedback enhance accessibility and usability.
- The modular architecture allows for future enhancements without overhauling the core design.

Future Scope:

- **Expand User Limit:** Modify file structure and encoding strategy to support more than 4 users.
- **Cloud-Based Monitoring:** Enable remote logging and notification of access events.
- **IoT Integration:** Integrate with mobile apps or voice assistants like Alexa/Google Assistant.
- **Advanced Security:** Add multi-factor authentication (e.g., face + PIN or OTP).
- **Environment Adaptation:** Enhance performance in low-light environments with IR cameras.

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